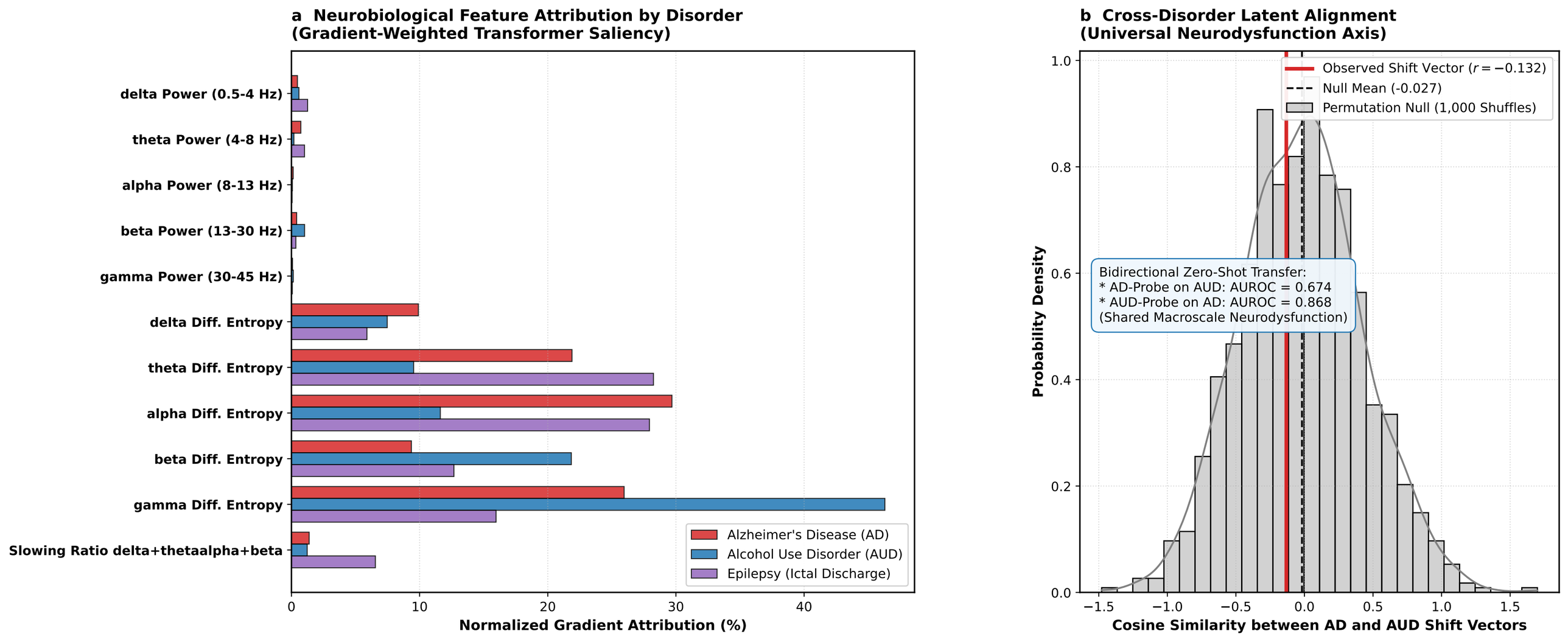
a Neurobiological Feature Attribution by Disorder
(Gradient-Weighted Transformer Saliency)
delta Power (0.5-4 Hz)
theta Power (4-8 Hz)
alpha Power (8-13 Hz)
beta Power (13-30 Hz)
gamma Power (30-45 Hz)
delta Diff. Entropy
theta Diff. Entropy
alpha Diff. Entropy
beta Diff. Entropy
gamma Diff. Entropy
Slowing Ratio delta+thetaalpha+beta
Alzheimer's Disease (AD)
Alcohol Use Disorder (AUD)
Epilepsy (Ictal Discharge)
0
10
20
30
40
Normalized Gradient Attribution (%)
b Cross-Disorder Latent Alignment
(Universal Neurodysfunction Axis)
Observed Shift Vector (r = −0.132)
Null Mean (-0.027)
Permutation Null (1,000 Shuffles)
Bidirectional Zero-Shot Transfer:
* AD-Probe on AUD: AUROC = 0.674
* AUD-Probe on AD: AUROC = 0.868
(Shared Macroscale Neurodysfunction)
Probability Density
0.0
0.2
0.4
0.6
0.8
1.0
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5
Cosine Similarity between AD and AUD Shift Vectors